

AI Social Media Analytics Framework

System Architecture & Technical Approach

National Technical Research Organisation (NTRO) · Theme: Blockchain & Cybersecurity

WHAT WE ARE BUILDING

An AI framework that ingests live social-media data across platforms and, on one timeline, infers how the audience feels, who they are, what is trending, and who influences whom.

Input: live posts, comments, interactions --> Output: audience-intelligence dashboard

The Five Core Components



Data Collection & Timeline

Multi-platform ingest, time-stamped store



Sentiment Inference

Nuanced emotion incl. sarcasm, over time



Demographic Profiling

Aggregate, anonymized: age / geo / language



Trend & Topic Detection

Rank & predict rising narratives

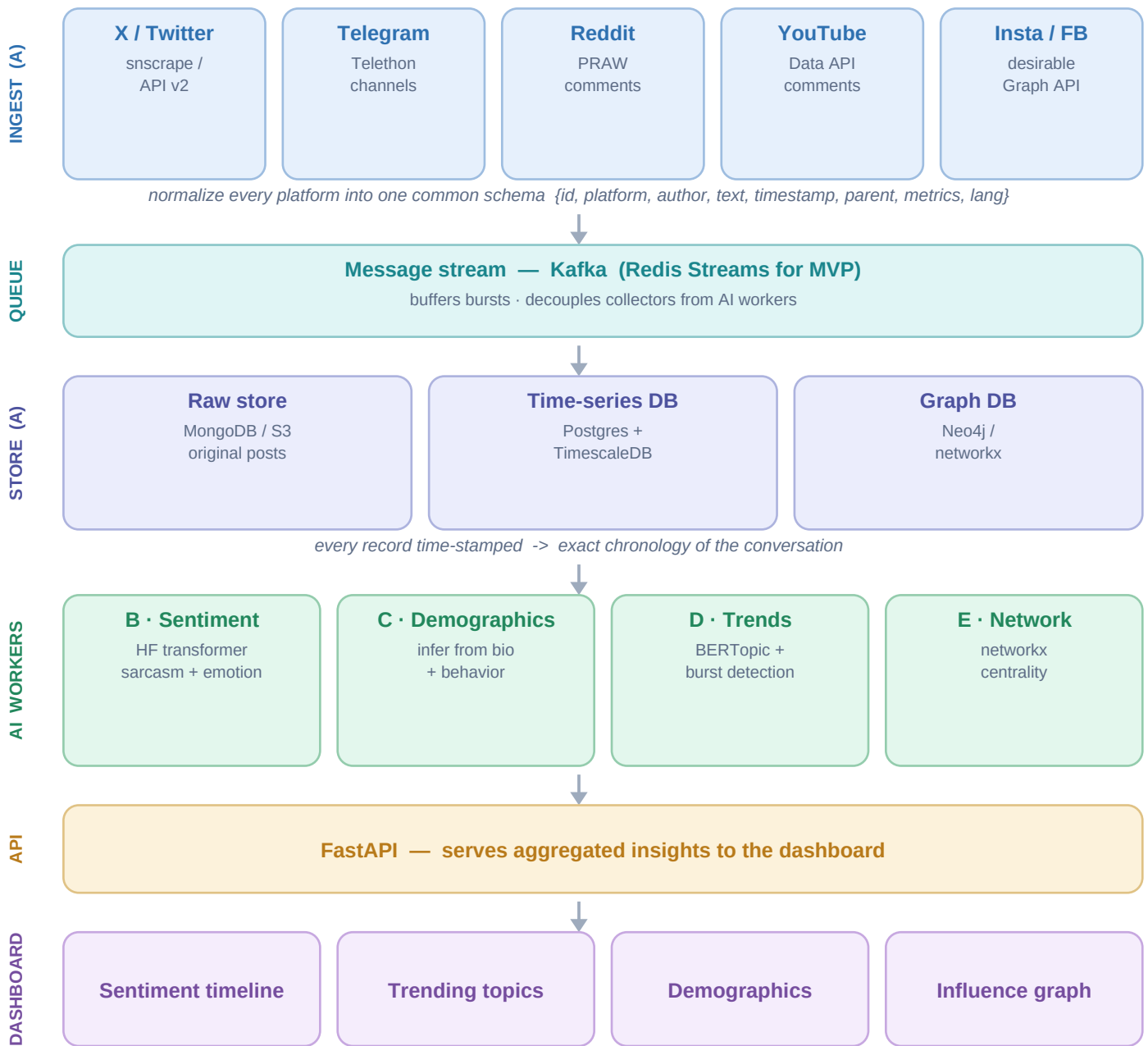


Link & Network Analysis

Map influence, find KOLs, track spread

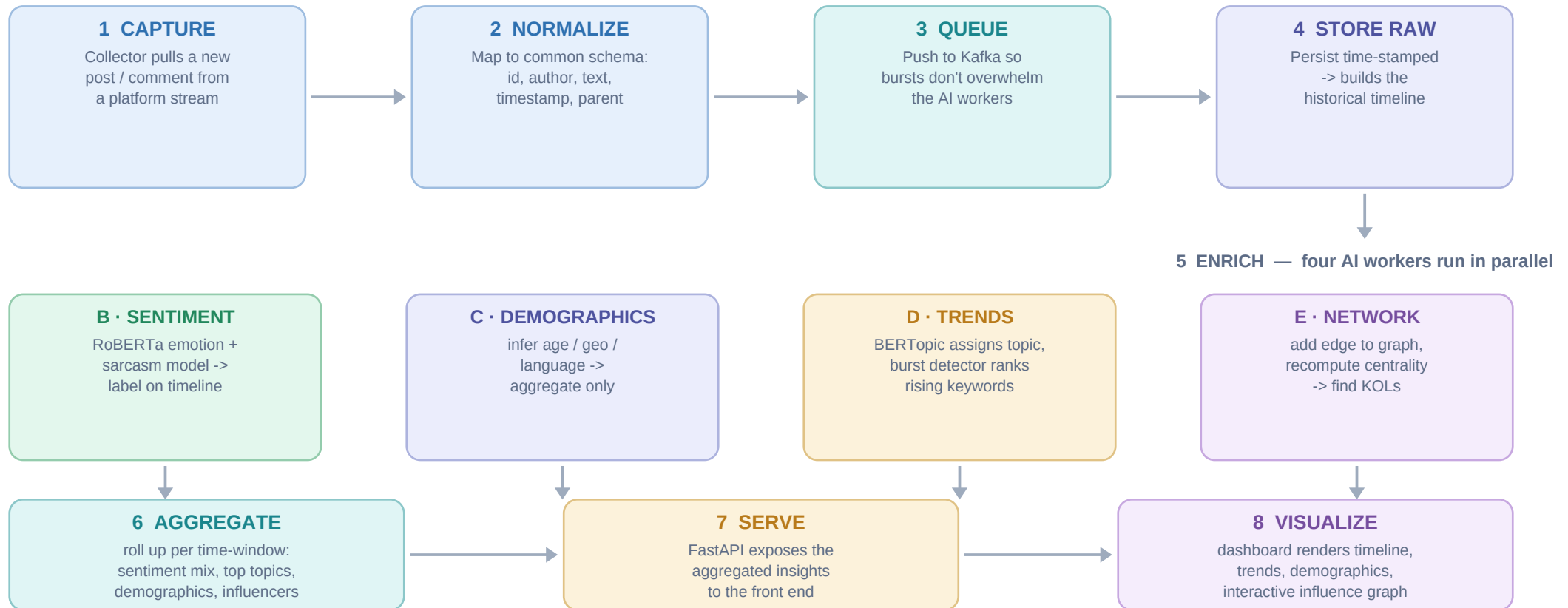
System Architecture

Six layers, top to bottom. Data flows down; each layer is independent and swappable.



Technical Approach — Life of a Post

Follow one post from capture to insight. Drop this flow straight into the presentation.



Reference — Stack, Data & Scope

TECHNOLOGY STACK

Collectors	Telethon (TG), PRAW (Reddit), YouTube Data API, snsrape/X API
Queue / stream	Kafka (Redis Streams for the MVP)
Databases	Postgres + TimescaleDB, MongoDB, Neo4j
Sentiment / NLP	HuggingFace transformers (RoBERTa emotion, sarcasm model)
Trends / topics	BERTopic, spaCy, burst detection
Network analysis	networkx / Neo4j — centrality, Louvain communities
API	FastAPI
Dashboard	React + D3 / PyVis or Streamlit (fastest)
Orchestration	Docker Compose, Celery / Airflow for scheduled jobs

DATA MODEL — WHY THREE STORES

Raw store	original posts & comments -> audit + reprocessing
Time-series DB	enriched metrics over time -> fast 'last 24h' queries
Graph DB	users = nodes, interactions = edges -> influence & spread

SCOPE — MVP VS FULL BUILD

MVP (always works)	Telegram + Reddit -> SQLite -> HF sentiment + BERTopic + networkx -> Streamlit dashboard. One machine, no Kafka.
Full build	add X / YouTube / Insta, Kafka, TimescaleDB, Neo4j, FastAPI + React, continuous live streaming.

TWO DECISIONS THAT WIN POINTS

1. Time-stamp everything	chronology is stressed in A, B, D, E — make it every panel's axis
2. Privacy by design	demographics aggregate & anonymized (@node_id) — the ethics answer

DATA-ACCESS NOTE

X / Twitter API is now paid & throttled. Lead the live demo with Telegram + Reddit + YouTube (all free official APIs); use limited / archived data for X. Have this answer ready for judges.